Seonghun Jung

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Research Interests

My research lies at the intersection of computer systems and AI. I study how emerging AI workloads challenge traditional system abstractions with their dynamic execution patterns and diverse resource demands. In particular, I focus on rethinking runtime trade-offs in interactive AI systems, where decisions about timing, resource allocation, and hardware placement can affect application-level outcomes. Guided by this perspective, my long-term goal is to design runtimes and resource management mechanisms that adapt AI execution to application requirements and hardware characteristics within heterogeneous computing systems.

Education

Korea Advanced Institute of Science and Technology (KAIST) **Sep. 2025 – Expected Aug. 2027**
Master of Science in Computer Science, School of Computing *Daejeon, Republic of Korea*

- Overall GPA 4.2/4.3

Sungkyunkwan University (SKKU) **Mar. 2020 – Aug. 2025**
Bachelor of Science in Computer Science and Engineering *Suwon, Republic of Korea*

- Overall GPA 4.19/4.5 (Magna Cum Laude)
- Leave of absence for mandatory military service (Jan. 2021 – Jul. 2022)

Publications

[ASPLOS] **Seonghun Jung**, Sieun Moon, Jiyoung Jeong, Jimin Lee, and Jaehyuk Huh, **"Beyond LLM Serving: Characterizing Vision-Language-Action Workloads for Embodied AI System Design"**, the International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), April, 2027 [paper] (Acceptance rate: 10.5%)

Research Experience

Characterizing VLA Workloads for Embodied AI System Design **Sep. 2025 – Present**
Graduate Research Assistant, advised by Prof. Jaehyuk Huh (CASYS), KAIST *Daejeon, Republic of Korea*

- Led the study from initial research-question formulation through experimental design, implementation, and manuscript writing.
- Characterized **architecture-dependent execution bottlenecks** across diverse VLA paradigms on server and edge GPUs (RTX 4090, Jetson Thor/Orin) using roofline analysis, revealing how action generation paradigms govern compute and memory demands.
- Designed a **phase-aware DVFS policy** that reduces latency by up to 18.8% and energy by 17.9% per inference, and built a MuJoCo-based **closed-loop evaluation harness** to quantify deployment trade-offs among execution speed, energy consumption, and task success.
- Derived design implications for **hardware architectures and serving-system abstractions** from the trade-offs driven by action-generation paradigms and closed-loop requirements.

ProfilingSense: Non-Intrusive Kubernetes Scheduler for Distributed DL Training **Sep. 2023 – Aug. 2024**
Undergraduate Research Assistant, advised by Prof. Euiseong Seo (CSL), SKKU *Suwon, Republic of Korea*

- Designed and implemented a **non-intrusive Kubernetes custom operator** that predicts job throughput from small-scale profiling in a dedicated GPU pool, requiring only a single YAML label and no code instrumentation.
- Designed a **history-free, efficiency-aware gang scheduling** policy combining predicted throughput per requested resource with wait-time aging to reorder jobs while preserving their resources during the main training run.
- Validated the operator on a **GCP testbed with 8 NVIDIA L4 GPUs** and extended an open-source cluster simulator to model our policy and baselines; replaying 2,312 Microsoft Philly jobs on an 864-GPU cluster cut average JCT by **22.1%** over the default scheduler and reduced average/P99 JCT by **4.5%/5.8%** and peak queue length by **8.2%** over FIFO-Gang.

Work Experience

Samsung Electronics | AI Center **Hwaseong, Republic of Korea**
Software Engineer Intern *Jan. 2025 – Feb. 2025*

- Diagnosed end-to-end service latency bottlenecks via Elastic APM/RUM distributed tracing and implemented a **hot-path caching layer** for frequently accessed views, reducing page-load latency by **26.5%**.

MLOps Engineer Intern *Sep. 2024 – Dec. 2024*

- Architected a **multi-tenant control plane** for a distributed Ray/Kubeflow hyperparameter-optimization platform serving 100+ engineers, integrating Volcano gang GPU scheduling, Redis state tracking, and Keycloak JWT authentication.
- Automated Kubernetes cluster deployment and observability pipelines using ArgoCD, Helm, and Istio service mesh, with centralized log aggregation via FluentBit and OpenSearch.

Projects

JOS: Exokernel-Style x86-64 OS **Spring 2026**
Course Project, Graduate Operating Systems, KAIST

- Built a 64-bit exokernel from boot: 4-level paging, ELF loading, trap and syscall handling, and SMP bring-up.
- Implemented preemptive scheduling, copy-on-write `fork` via page-fault upcalls, IPC, and a user-level file system.

xv6 Kernel Extensions **Fall 2023**
Course Project, Operating Systems, SKKU

- Replaced round-robin with a Linux-style CFS: nice-to-weight table, weighted timeslices, sleeper fairness, 64-bit `vruntime`.
- Added `mmap` with page-fault-driven demand paging and Clock (second-chance) page replacement with swap space.

Selected Seminars

KV-Cache Management: From Heterogeneous LLMs to Embodied VLAs **Fall 2025 – Spring 2026**
MLSys Seminars, Computer Architecture and Systems Lab (CASYS), KAIST

- **Multi-task VLA serving:** Analyzed cross-task KV-cache sharing and lifecycle management in embodied control loops, based on *"OxyGen: Unified KV Cache Management for VLA Inference under Multi-Task Parallelism"* (Li et al., arXiv 2026) [**Slides**]
- **Heterogeneous LLM serving:** Analyzed two-level memory allocation for mitigating fragmentation across heterogeneous attention layers, based on *"Jenga: Effective Memory Management for Serving LLM with Heterogeneity"* (Zhang et al., SOSP 2025) [**Slides**]

OS Virtual Memory Abstractions for High-Throughput LLM Serving **Spring 2026**
Graduate Operating Systems Course Seminar, School of Computing, KAIST

- **Foundational LLM serving:** Co-presented an analysis of block-level paging and copy-on-write KV-cache sharing (with Changyong Eom), based on *"Efficient Memory Management for Large Language Model Serving with PagedAttention"* (Kwon et al., SOSP 2023) [**Slides**]

Teaching & Mentoring

Teaching Assistant, Computer Organization **Spring 2026**
School of Computing, KAIST – Outstanding Teaching Assistant Award

- **MMU Simulator Lab**: Designed the assignment on virtual-to-physical address translation (page-table walk and TLB), led Q&A sessions, and managed grading for about 160 international students.

Teaching Assistant, System Programming **Fall 2025**
School of Computing, KAIST – Outstanding Teaching Assistant Award

- **Data Lab (Manipulating Bits)**: Prepared bit-level integer and IEEE-754 floating-point puzzles and autograders in C, ran Q&A on Piazza, and managed grading for 180+ international students.

Volunteer Instructor, SKKU Software Mentoring Program **May 2024 – Sep. 2024**
Suwon Girls' High School *Suwon, Republic of Korea*

- Taught Python fundamentals and introductory deep-learning concepts, and mentored student team projects.

Undergraduate Teaching Assistant, Engineering Computer Programming **Fall 2023**
Department of Computer Science and Engineering, SKKU

- Led lab sessions on C++ fundamentals and data structures for about 80 international students.

Honors & Awards

Scholarships

National Full Graduate Scholarship, Korean Government (M.S. full tuition, 4 semesters) **Sep. 2025 – Aug. 2027**
Sungkyun Software and Research Promotion Scholarship, SKKU (B.S. full tuition, 8 semesters) **Mar. 2020 – Aug. 2025**

Awards

Outstanding Teaching Assistant Award, School of Computing, KAIST **Dec. 2025, Jul. 2026**
Gold Prize (2nd of 16 Teams), SKKU Hackathon — Backend Developer & Team Lead [**Code**] **Sep. 2023**
Top 15 of 70+ Participants, SKKU Programming Competition **May 2024, Jun. 2025**

Certifications

Certified Kubernetes Administrator (CKA), The Linux Foundation **Feb. 2025**
S/W Certification Professional, Samsung Electronics DX **Oct. 2024**

Skills

Programming: Python, C/C++, CUDA
Systems & Tools: PyTorch, vLLM, Ray, Kubernetes, Kubeflow, Horovod, Docker, Linux, Nsight Compute/Systems (NCU/NSYS), MuJoCo, Isaac Sim
Languages: Korean (native), English (TOEFL iBT 5.0/6.0; equivalent to 100/120 on the previous scale)